

TLQLDMR: A Transfer Learning Quantile Regression Model for Wind Power Forecasting under Extreme Weather Conditions

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ABSTRACT

Accurate wind power forecasting under extreme weather conditions is critical for ensuring power grid stability, yet remains fundamentally challenging due to the inherent scarcity of extreme event samples and the substantial distribution shift between normal and anomalous meteorological regimes. Conventional data-driven approaches, predicated on the independent and identically distributed (i.i.d.) assumption, exhibit significant performance degradation when confronted with such sporadic high-impact events. Moreover, existing transfer learning methodologies predominantly emphasize unsupervised feature alignment while overlooking the discriminative information embedded in scarce target labels, and standard large margin regression techniques impose symmetric error distributions that are inherently incompatible with the asymmetric nature of probabilistic interval forecasting.

To address these limitations, we propose a novel Transfer Learning Quantile Large Margin Distribution Machine for Regression (TLQLDMR). Our approach introduces three key innovations: (1) an Asymmetric Variance Regularizer that theoretically reconciles the tension between distributional robustness and quantile estimation by compressing error dispersion while preserving the requisite skewness for accurate quantile prediction; (2) a Supervised Transfer Learning framework incorporating dual-space adaptation, which combines Maximum Mean Discrepancy (MMD) minimization in the feature space with a Weighted Pinball Loss mechanism in the label space, strategically leveraging scarce extreme weather samples as high-weight corrective anchors; and (3) a scalable optimization algorithm based on Nyström approximation and stochastic gradient descent that strictly reduces computational complexity from $O(N^3)$ to a manageable $O(T \cdot N \cdot m)$, enabling efficient training on large-scale wind farm datasets under dual seconds. Extensive empirical studies—including comprehensive ablation and varying signal-to-noise ratio analyses—demonstrate that TLQLDMR consistently outperforms state-of-the-art baselines. Crucially, the model yields substantially tighter deterministic accuracy and superior probabilistic interval performance, reliably conquering the nominal 95% prediction interval coverage benchmark amidst severe meteorological shifts and measurement noise.

1. Introduction

2. Related Work

This section briefly reviews the theoretical foundations related to the proposed method, including Support Vector Quantile Regression, Large Margin Distribution Machine for Regression, and Transfer Learning with Maximum Mean Discrepancy.

2.1. Support Vector Quantile Regression (SVQR)

Traditional Support Vector Regression (SVR) relies on the ϵ -insensitive loss function, which assumes a symmetric error distribution (e.g., Gaussian) and estimates the conditional mean of the target variable. However, in scenarios like partial shading or wind curtailment, the noise distribution is often asymmetric and heteroscedastic. To address this, Support Vector Quantile Regression (SVQR) was introduced to estimate conditional quantiles, providing a comprehensive view of the outcome distribution.

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The core of SVQR is the non-symmetric pinball loss function, defined as:

$$L_{\tau}(u) = \begin{cases} \tau u, & u \geq 0 \\ (\tau - 1)u, & u < 0, \end{cases} \quad (1)$$

where $\tau \in (0, 1)$ indicates the target quantile. Unlike SVR, SVQR finds a decision boundary that divides the data such that a proportion τ of the samples lie below the boundary. The optimization problem minimizes the structural risk along with the empirical risk measured by the pinball loss:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n L_{\tau}(y_i - (\mathbf{w}^T \phi(\mathbf{x}_i) + b)), \quad (2)$$

this formulation allows SVQR to capture heavy-tailed distributions and outliers more effectively than mean-based regression.

2.2. Large Margin Distribution Machine for Regression (LDMR)

While SVR and SVQR focus on maximizing the minimum margin (the distance from the closest instances to the hyperplane), they largely ignore the statistical distribution of the margin over the entire dataset. The **Large Margin Distribution Machine (LDMR)** was proposed to improve generalization by explicitly optimizing the distribution of the margins (or residuals).

The key idea of LDMR is to not only maximize the margin of the support vectors but also to minimize the variance of the error distribution for all samples. A typical LDMR formulation for regression integrates two additional statistical regularization terms into the objective function:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + \lambda_1 \text{Var}(\text{Residuals}) + \lambda_2 \text{Mean}(\text{Residuals}) + C \sum_{i=1}^n L(y_i, f(\mathbf{x}_i)), \quad (3)$$

by constraining the variance, LDMR forces the prediction errors to be clustered tightly, leading to a more compact and robust hypothesis space. Our work extends this concept by modifying the symmetric variance term of standard LDMR to an asymmetric one suitable for quantile regression.

2.3. Transfer Learning and Maximum Mean Discrepancy (MMD)

Standard machine learning models operate under the assumption that training (source) and testing (target) data are drawn from the same probability distribution. When this assumption is violated—such as in wind power forecasting across different weather regimes—the model performance degrades significantly. Transfer learning aims to transfer knowledge from a label-rich source domain to a target domain with different but related distributions.

A fundamental challenge in transfer learning is measuring and reducing the distribution discrepancy. Maximum Mean Discrepancy (MMD) is a widely used non-parametric metric that measures the distance between two distributions in a Reproducing Kernel Hilbert Space (RKHS). Given source data $\{\mathbf{x}_{S,i}\}$ and target data $\{\mathbf{x}_{T,j}\}$, the squared MMD distance is defined as:

$$\text{MMD}^2(D_S, D_T) = \left\| \frac{1}{n_S} \sum_{i=1}^{n_S} \phi(\mathbf{x}_{S,i}) - \frac{1}{n_T} \sum_{j=1}^{n_T} \phi(\mathbf{x}_{T,j}) \right\|_{\mathcal{H}}^2, \quad (4)$$

where $\phi(\cdot)$ is the feature mapping induced by the kernel. By minimizing the MMD distance during training, the model learns a feature representation where the source and target domains are statistically aligned, enabling the classifier or regressor trained on the source domain to generalize well to the target domain.

3. Proposed Methodology

In this section, we elaborate on the construction of the Transfer Learning Quantile Regression Model (TLQLDMR). We first define the necessary notations, then present the final objective function which integrates Structural Risk, Distribution Variance, Domain Adaptation (MMD), and Weighted Pinball Loss. Finally, we describe the constraints introduced to handle the non-differentiability of the Quantile Loss.

3.1. Notations

Consider the problem of wind power forecasting under extreme weather conditions. We define the **Source Domain (Normal Weather)** as $D_S = \{(\mathbf{x}_{S,i}, y_{S,i})\}_{i=1}^{n_S}$, containing abundant conventional weather samples, and the **Target Domain (Extreme Weather)** as $D_T = \{(\mathbf{x}_{T,j}, y_{T,j})\}_{j=1}^{n_T}$, containing scarce extreme weather samples. The input space $\mathcal{X} \subseteq \mathbb{R}^d$ represents feature vectors constructed via sliding window processing, and the output space $\mathcal{Y} \subseteq \mathbb{R}$ represents wind power. The sample sizes satisfy $n_T \ll n_S$.

Due to drastic meteorological changes during extreme events, the joint distributions differ significantly: $P_S(\mathbf{X}_S, \mathbf{Y}_S) \neq P_T(\mathbf{X}_T, \mathbf{Y}_T)$. This distribution discrepancy implies that a model trained solely on P_S will fail on P_T . We define a feature mapping $\phi : \mathcal{X} \rightarrow \mathcal{H}$ that maps the input to a high-dimensional Reproducing Kernel Hilbert Space (RKHS). The prediction model is defined as:

$$f(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b, \quad (5)$$

where $\mathbf{w}$ is the weight vector and b is the bias. The quantile $\tau \in (0, 1)$ is set to a specific value (e.g., $\tau = 0.95$) for interval prediction (e.g., predicting the upper bound).

3.2. TLQLDMR Model Formulation

The overall architecture of the proposed TLQLDMR model is illustrated in Fig. 1. The model integrates four key components: structural risk minimization, asymmetric variance regularization, MMD-based domain adaptation, and weighted pinball loss for supervised transfer learning.

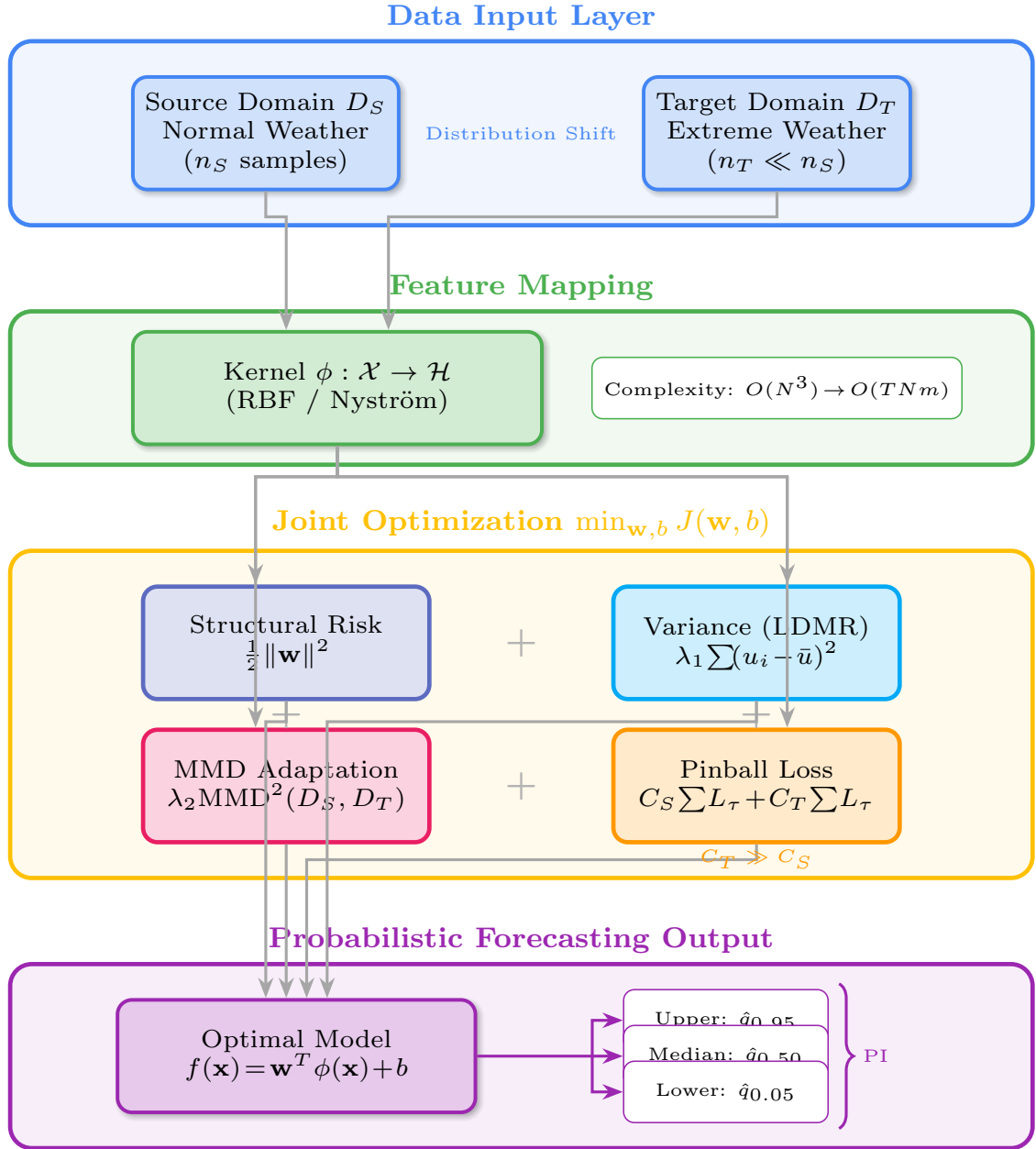


Fig. 1. The overall framework of the proposed TLQLDMR model. Source domain (normal weather) and target domain (extreme weather) data are mapped into a high-dimensional RKHS via kernel mapping. The joint optimization objective integrates structural risk, distribution variance, MMD-based domain adaptation, and weighted pinball loss. The model outputs probabilistic prediction intervals for robust wind power forecasting under extreme conditions.

3.2.1. Objective Function

Our optimization goal is to find the optimal $\mathbf{w}, b$ and slack variables ξ, ξ^* to minimize the following objective function J . Let $\bar{u}_S = \frac{1}{n_S} \sum_{i=1}^{n_S} (y_{S,i} - f(\mathbf{x}_{S,i}))$ denote the mean residual over the source domain:

$$\begin{aligned}
 \min_{\mathbf{w}, b, \xi, \xi^*} J = & \underbrace{\frac{1}{2} \|\mathbf{w}\|^2}_{(1) \text{ Structural Risk}} \\
 & + \underbrace{\lambda_1 \sum_{i=1}^{n_S} ((y_{S,i} - \mathbf{w}^T \phi(\mathbf{x}_{S,i}) - b) - \bar{u}_S)^2}_{(2) \text{ Distribution Variance (LDMR Core)}} \\
 & + \underbrace{\lambda_2 \left\| \frac{1}{n_S} \sum_{i=1}^{n_S} \phi(\mathbf{x}_{S,i}) - \frac{1}{n_T} \sum_{j=1}^{n_T} \phi(\mathbf{x}_{T,j}) \right\|_H^2}_{(3) \text{ Domain Adaptation (MMD)}} \\
 & + \underbrace{C_S \sum_{i=1}^{n_S} L_\tau(y_{S,i}, f(\mathbf{x}_{S,i})) + C_T \sum_{j=1}^{n_T} L_\tau(y_{T,j}, f(\mathbf{x}_{T,j}))}_{(4) \text{ Weighted Pinball Loss}},
 \end{aligned} \tag{6}$$

the physical meaning of each term is as follows:

1. **Structural Risk:** This term controls the complexity of the model and prevents overfitting, embodying the maximum margin principle of SVM.
2. **Distribution Variance:** This term is the core of our proposed Quantile LDMR (QLDMR), incorporating a critical modification to the original LDMR variance.
 - **Theoretical Conflict:** The original LDMR minimizes the variance of absolute residuals, which assumes a symmetric error distribution suitable for mean regression. However, this symmetry contradicts the inherent asymmetry of Quantile Regression (governed by τ). Applying a symmetric variance penalty would force the residual distribution towards the center, degrading the skewness required for accurate quantile estimation.
 - **Asymmetric Modification:** To resolve this, we employ an **Asymmetric Variance** regularization based on the algebraic (signed) residuals. This design allows the model to reduce error dispersion (compactness) without constraining the distribution's direction, thereby preserving the non-symmetric properties essential for the Pinball Loss and improving interval coverage.
3. **Domain Adaptation (MMD):** In the context of wind power forecasting under extreme weather conditions, our primary objective is to mitigate the differences in data distributions across various domains by means of learning a feature transformation. Consequently, the effective measurement of distribution disparities between domains becomes a paramount concern. In this pursuit, it is crucial to select a method suitable for quantifying distribution dissimilarity, while ensuring that the chosen method is computationally efficient and easily integrated into the optimization problem of our model. Maximum Mean Discrepancy (MMD) has been demonstrated as an effective approach for estimating the distance between two distributions, offering a relatively straightforward implementation with fast computation speed. It targets the input space $\mathcal{X}$ to minimize the distance between the distribution centers of the source and target domains.
4. **Weighted Pinball Loss:** This term targets the output space $\mathcal{Y}$ and acts as the decisive component for ‘‘Supervised Transfer Learning’’. Unlike unsupervised transfer which relies solely on feature alignment, we explicitly incorporate the target domain samples D_T into the loss calculation for the following reasons:
 - **Corrective Anchors:** The target samples act as ‘‘anchors for correction’’. The source domain data (D_S) teaches the model general laws (e.g., higher wind speed leads to higher power), but extreme weather samples (D_T) often violate these laws (e.g., high wind speed causing cut-out and zero power). If D_T were excluded from the loss, the model, driven implicitly by D_S , would wrongly predict full power during

extreme events. Including D_T in the loss imposes a heavy penalty for such mispredictions, forcing the model to learn the specific mapping of extreme events.

- **Differentiation from MMD:** While MMD brings the “appearance” (features) of normal and extreme weather closer, the Pinball Loss ensures the “content” (labels) is accurate. It forces the regression curve to correctly envelope the rare data points that MMD cannot correct on its own.
- **Asymmetric Weighted Strategy:** We employ a weighted strategy to address the extreme sample imbalance ($n_T \ll n_S$). By setting $C_T \gg C_S$ (e.g., $C_T = 100C_S$), the optimization process assigns substantially higher importance to target domain samples. This weighting scheme ensures that prediction errors on scarce extreme weather data incur penalties commensurate with their practical significance, preventing them from being dominated by the abundant source domain samples during gradient updates.

3.2.2. Constraints

To handle the non-differentiability of the Pinball Loss, we introduce slack variables ξ_k, ξ_k^* to transform it into linear constraints. For every sample k in the training set $D = D_S \cup D_T$:

$$\begin{cases} y_k - (\mathbf{w}^T \phi(\mathbf{x}_k) + b) \leq \xi_k, & \forall k \in \{1, \dots, n_S + n_T\} \\ (\mathbf{w}^T \phi(\mathbf{x}_k) + b) - y_k \leq \xi_k^*, & \forall k \in \{1, \dots, n_S + n_T\} \\ \xi_k \geq 0, \quad \xi_k^* \geq 0, \end{cases} \quad (7)$$

in the objective function, the term $L_\tau(y_k, f(\mathbf{x}_k))$ corresponds to $\tau \xi_k + (1 - \tau) \xi_k^*$.

3.3. Model Optimization

In this section, we derive the dual problem of TLQLDMR to enable efficient solving using Quadratic Programming (QP).

3.3.1. Matrix Reformulation of the Primal Problem

To facilitate the derivation, we rewrite the variance term and the MMD term in the objective function [Eq. (6)] using matrix traces or norms. Let $X = [X_S, X_T]$ denote the entire training set with $N = n_S + n_T$ samples. For consistency with standard derivations, we add a coefficient of 1/2 to the variance and MMD terms. The primal objective can be reformulated as:

$$\min_{\mathbf{w}, b, \xi, \xi^*} \frac{1}{2} \|\mathbf{w}\|^2 + \frac{\lambda_1}{2} \text{Var}_S(\mathbf{w}, b) + \frac{\lambda_2}{2} \text{MMD}^2(\mathbf{w}) + \sum_{k=1}^N C_k L_\tau(y_k, f(\mathbf{x}_k)) \quad (8)$$

where C_k takes value C_S if sample $k \in D_S$ and C_T if $k \in D_T$.

We introduce two auxiliary matrices to express the variance and MMD terms in Eq. (8) as quadratic forms of the weights:

1. **Variance Matrix S:** To represent the source domain variance $\sum_{i \in D_S} (y_{S,i} - f(\mathbf{x}_{S,i}) - \bar{u}_S)^2$, we define a centering matrix $\mathbf{H}_S = \mathbf{I}_{n_S} - \frac{1}{n_S} \mathbf{1}_{n_S} \mathbf{1}_{n_S}^T$. The variance term can be written as $\|\mathbf{H}_S(\mathbf{y}_S - \mathbf{f}_S)\|^2$. To extend this to the full dataset, we define an $N \times N$ matrix $\mathbf{S}$:

$$\mathbf{S} = \begin{bmatrix} \mathbf{H}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (9)$$

2. **MMD Matrix M:** The MMD term represents the distance between means. We define the MMD coefficient matrix $\mathbf{M} \in \mathbb{R}^{N \times N}$ as:

$$M_{ij} = \begin{cases} \frac{1}{n_S^2} & i, j \in D_S \\ \frac{1}{n_T^2} & i, j \in D_T \\ \frac{-1}{n_S n_T} & \text{otherwise} \end{cases} \quad (10)$$

3.3.2. Representer Theorem and Kernelization

According to the Representer Theorem [Belkin et al. \(2006\)](#), the optimal weight vector $\mathbf{w}$ lies in the span of the feature mappings of the training samples:

$$\mathbf{w} = \sum_{i=1}^N \beta_i \phi(\mathbf{x}_i) = \Phi \boldsymbol{\beta} \quad (11)$$

where $\Phi = [\phi(\mathbf{x}_1), \dots, \phi(\mathbf{x}_N)]$ is the feature matrix and $\boldsymbol{\beta} = [\beta_1, \dots, \beta_N]^T$ are the coefficients. Substituting Eq. (11) into the prediction function, we get $f(\mathbf{X}) = \boldsymbol{\beta}^T \mathbf{K} + \mathbf{b} \mathbf{1}^T$, where $\mathbf{K} = \Phi^T \Phi$ is the $N \times N$ kernel matrix. Consequently, using Eq. (9) and Eq. (10), the structural risk, variance, and MMD terms can be expressed in terms of $\boldsymbol{\beta}$:

- Structural Risk: $\|\mathbf{w}\|^2 = \boldsymbol{\beta}^T \mathbf{K} \boldsymbol{\beta}$.
- Distribution Variance: The quadratic part with respect to $\boldsymbol{\beta}$ is $\boldsymbol{\beta}^T \mathbf{K} \mathbf{S} \mathbf{K} \boldsymbol{\beta}$. Note that the bias b is eliminated by the centering matrix $\mathbf{H}_S$.
- Domain Adaptation (MMD): $\text{MMD}^2(\mathbf{w}) = \boldsymbol{\beta}^T \mathbf{K} \mathbf{M} \mathbf{K} \boldsymbol{\beta}$.

Combining these, we define the total coefficient matrix $\boldsymbol{\Omega}$:

$$\boldsymbol{\Omega} = \mathbf{K} + \lambda_1 \mathbf{K} \mathbf{S} \mathbf{K} + \lambda_2 \mathbf{K} \mathbf{M} \mathbf{K} \quad (12)$$

By substituting the kernelized terms and Eq. (12) into Eq. (8), the kernelized primal problem becomes (ignoring constant terms w.r.t b in variance):

$$\min_{\boldsymbol{\beta}, b, \xi, \xi^*} \quad \frac{1}{2} \boldsymbol{\beta}^T \boldsymbol{\Omega} \boldsymbol{\beta} - (\lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y})^T \boldsymbol{\beta} + \sum_{k=1}^N C_k (\tau \xi_k + (1 - \tau) \xi_k^*) \quad (13)$$

subject to the constraints from Eq. (7):

$$y_k - (\mathbf{K}_k \boldsymbol{\beta} + b) \leq \xi_k, \quad (\mathbf{K}_k \boldsymbol{\beta} + b) - y_k \leq \xi_k^*, \quad \xi_k, \xi_k^* \geq 0 \quad (14)$$

3.3.3. Lagrangian and Dual Problem

We verify the problem by introducing Lagrange multipliers $\alpha_k, \alpha_k^*, \mu_k, \mu_k^* \geq 0$ for Eq. (14). The Lagrangian function $\mathcal{L}$ is constructed as:

$$\begin{aligned} \mathcal{L} = & \frac{1}{2} \boldsymbol{\beta}^T \boldsymbol{\Omega} \boldsymbol{\beta} - (\lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y})^T \boldsymbol{\beta} + \sum_{k=1}^N C_k (\tau \xi_k + (1 - \tau) \xi_k^*) \\ & - \sum_{k=1}^N \alpha_k (\xi_k - y_k + \mathbf{K}_k \boldsymbol{\beta} + b) - \sum_{k=1}^N \alpha_k^* (\xi_k^* + y_k - \mathbf{K}_k \boldsymbol{\beta} - b) - \sum (\mu_k \xi_k + \mu_k^* \xi_k^*) \end{aligned} \quad (15)$$

By setting the partial derivatives of Eq. (15) with respect to the primal variables $(b, \xi, \xi^*, \boldsymbol{\beta})$ to zero, we obtain the KKT conditions:

$$\frac{\partial \mathcal{L}}{\partial b} = 0 \implies \sum_{k=1}^N (\alpha_k - \alpha_k^*) = 0 \quad (16)$$

$$\frac{\partial \mathcal{L}}{\partial \xi_k} = 0, \quad \frac{\partial \mathcal{L}}{\partial \xi_k^*} = 0 \implies \alpha_k \in [0, C_k \tau], \quad \alpha_k^* \in [0, C_k (1 - \tau)] \quad (17)$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\beta}} = 0 \implies \boldsymbol{\Omega} \boldsymbol{\beta} - \lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y} - \mathbf{K} (\boldsymbol{\alpha} - \boldsymbol{\alpha}^*) = 0 \quad (18)$$

Let $\boldsymbol{\gamma} = \boldsymbol{\alpha} - \boldsymbol{\alpha}^*$. From Eq. (18), the optimal $\boldsymbol{\beta}$ is derived as:

$$\boldsymbol{\beta} = \boldsymbol{\Omega}^{-1} (\mathbf{K} \boldsymbol{\gamma} + \lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y}) \quad (19)$$

Substituting Eq. (19) back into Eq. (15), and simplifying using Eq. (16) and Eq. (17), we arrive at the **Dual Problem**, which is a standard Quadratic Programming (QP) problem:

$$\begin{aligned}
\min_{\alpha, \alpha^*} \quad & \frac{1}{2}(\alpha - \alpha^*)^T \mathbf{Q}_{dual}(\alpha - \alpha^*) + \mathbf{p}_{dual}^T(\alpha - \alpha^*) \\
\text{s.t.} \quad & \sum_{k=1}^N (\alpha_k - \alpha_k^*) = 0 \\
& 0 \leq \alpha_k \leq C_k \tau, \quad \forall k \\
& 0 \leq \alpha_k^* \leq C_k(1 - \tau), \quad \forall k
\end{aligned} \tag{20}$$

where the dual matrix $\mathbf{Q}_{dual}$ and vector $\mathbf{p}_{dual}$ are defined as:

$$\mathbf{Q}_{dual} = \mathbf{K}\mathbf{\Omega}^{-1}\mathbf{K}, \quad \mathbf{p}_{dual} = \lambda_1 \mathbf{K}\mathbf{\Omega}^{-1}\mathbf{K}\mathbf{S}\mathbf{Y} - \mathbf{Y} \tag{21}$$

Once the optimal α, α^* are found using a QP solver for Eq. (20), we can recover β using Eq. (19) and construct the final decision function $f(\mathbf{x})$.

3.4. The Algorithm and Complexity Analysis

The complete training process of the proposed TLQLDMR model is summarized in Algorithm 1. The algorithm mainly consists of matrix construction and solving a Quadratic Programming (QP) problem.

Algorithm 1: TLQLDMR Training Algorithm

Input : Source Data $D_S = \{(\mathbf{x}_{S,i}, y_{S,i})\}_{i=1}^{n_S}$, Target Data $D_T = \{(\mathbf{x}_{T,j}, y_{T,j})\}_{j=1}^{n_T}$, Quantile τ , Kernel parameters, Regularization parameters λ_1, λ_2 , Pinball Loss weights C_S, C_T .
Output: Optimal dual variables α, α^* and bias b .
// Step 1: Data Preparation
1 Combine Source and Target data: $\mathbf{X} = [\mathbf{X}_S, \mathbf{X}_T]$, $\mathbf{Y} = [\mathbf{Y}_S; \mathbf{Y}_T]$, $N = n_S + n_T$;
2 Set weights vector $\mathbf{C} \in \mathbb{R}^N$ where $C_k = C_S$ for $k \leq n_S$ and C_T otherwise;
// Step 2: Matrix Construction
3 Compute Kernel Matrix $\mathbf{K} \in \mathbb{R}^{N \times N}$ using the specified kernel function;
4 Construct Variance Matrix $\mathbf{S}$ using Eq. (9);
5 Construct MMD Matrix $\mathbf{M}$ using Eq. (10);
// Step 3: Dual Problem Formulation
6 Compute coefficient matrix $\mathbf{\Omega} = \mathbf{K} + \lambda_1 \mathbf{K}\mathbf{S}\mathbf{K} + \lambda_2 \mathbf{K}\mathbf{M}\mathbf{K}$ using Eq. (12);
7 Invert $\mathbf{\Omega}$ to fit the dual form;
8 Compute Dual Matrix $\mathbf{Q}_{dual} = \mathbf{K}\mathbf{\Omega}^{-1}\mathbf{K}$ using Eq. (21);
9 Compute Dual Vector $\mathbf{p}_{dual} = \lambda_1 \mathbf{K}\mathbf{\Omega}^{-1}\mathbf{K}\mathbf{S}\mathbf{Y} - \mathbf{Y}$ using Eq. (21);
// Step 4: Solve QP
10 Solve the QP problem in Eq. (20) providing $\mathbf{Q}_{dual}$, $\mathbf{p}_{dual}$, and constraints $0 \leq \alpha_k \leq C_k \tau, 0 \leq \alpha_k^* \leq C_k(1 - \tau)$;
11 **Return** Optimal α, α^* ;
// Step 5: Recover Primal Variables
12 Compute $\gamma = \alpha - \alpha^*$;
13 Recover β using Eq. (19);
14 Compute bias b using complementary slackness conditions (averaged over Support Vectors);

The computational complexity of TLQLDMR primarily stems from three parts: matrix multiplication, matrix inversion, and the QP solver.

1. **Matrix Operations**: The construction of $\mathbf{K}$, $\mathbf{S}$, and $\mathbf{M}$ involves $O(N^2d)$ or $O(N^2)$ operations. The calculation of $\mathbf{\Omega}$ involves matrix multiplications of size $N \times N$, leading to a complexity of $O(N^3)$.
2. **Matrix Inversion**: Inverting the $N \times N$ matrix $\mathbf{\Omega}$ requires $O(N^3)$ complexity.

3. **QP Solver:** Solving the standard Quadratic Programming problem for the dual variables typically has a complexity ranging from $O(N^{2.3})$ to $O(N^3)$ depending on the solver implementation (e.g., Interior Point Method).

Therefore, the overall time complexity of the proposed algorithm is approximately $O(N^3)$. Given that $N = n_S + n_T$ and $n_T \ll n_S$, the scale is dominated by the source domain sample size. While this complexity is acceptable for moderate-sized datasets, it becomes prohibitive for large-scale wind power forecasting scenarios involving tens of thousands of samples. To overcome this computational bottleneck and enable practical deployment in real-world wind farm applications, we develop a scalable optimization algorithm in the following subsection.

3.5. Scalable Optimization via Nyström Approximation

To address the computational bottleneck of the standard QP-based solver, we propose a scalable optimization algorithm that combines Nyström approximation with stochastic gradient descent. This approach reduces the time complexity from $O(N^3)$ to $O(T \cdot N \cdot m)$, where T denotes the number of iterations and $m \ll N$ is the number of landmark points.

3.5.1. Nyström Feature Mapping

The core idea is to approximate the implicit infinite-dimensional feature mapping $\phi(\mathbf{x})$ with an explicit finite-dimensional representation. Given the training data, we first select m landmark points $Z = \{z_1, \dots, z_m\}$ from the source domain D_S using K-Means clustering to capture the underlying data manifold.

Let $\mathbf{W} \in \mathbb{R}^{m \times m}$ denote the kernel matrix of landmarks, where $W_{ij} = k(z_i, z_j)$. The explicit feature mapping $\psi : \mathcal{X} \rightarrow \mathbb{R}^m$ is then constructed as:

$$\psi(\mathbf{x}) = \mathbf{W}^{-1/2} [k(\mathbf{x}, z_1), \dots, k(\mathbf{x}, z_m)]^T \quad (22)$$

where $\mathbf{W}^{-1/2}$ is computed via eigendecomposition. This mapping satisfies the approximation property $k(\mathbf{x}_i, \mathbf{x}_j) \approx \psi(\mathbf{x}_i)^T \psi(\mathbf{x}_j)$, enabling us to work directly in the primal space.

3.5.2. Primal Formulation with Explicit Features

With the explicit feature mapping, the prediction function becomes $f(\mathbf{x}) = \mathbf{w}^T \psi(\mathbf{x}) + b$, where $\mathbf{w} \in \mathbb{R}^m$. Let $\Psi_S \in \mathbb{R}^{n_S \times m}$ and $\Psi_T \in \mathbb{R}^{n_T \times m}$ denote the mapped feature matrices for source and target domains, respectively. The regularization matrix $\mathbf{R} \in \mathbb{R}^{m \times m}$ incorporating structural risk, variance, and MMD terms is formulated as:

$$\mathbf{R} = \mathbf{I} + \lambda_1 \underbrace{\Psi_S^T \mathbf{H}_S \Psi_S}_{\text{Variance}} + \lambda_2 \underbrace{(\bar{\Psi}_S - \bar{\Psi}_T)(\bar{\Psi}_S - \bar{\Psi}_T)^T}_{\text{MMD}} \quad (23)$$

where $\mathbf{H}_S = \mathbf{I}_{n_S} - \frac{1}{n_S} \mathbf{1}_{n_S} \mathbf{1}_{n_S}^T$ is the centering matrix, and $\bar{\Psi}_S, \bar{\Psi}_T$ are the mean feature vectors of source and target domains.

3.5.3. Stochastic Gradient Descent Optimization

Instead of solving the dual QP problem, we directly optimize the primal objective using mini-batch stochastic gradient descent with the Adam optimizer. For each sample k in a mini-batch $\mathcal{B}$, the sub-gradient of the Pinball Loss with respect to the prediction is:

$$g_k = \frac{\partial L_\tau}{\partial \hat{y}_k} = \begin{cases} \tau - 1, & \text{if } y_k < \hat{y}_k \\ \tau, & \text{if } y_k \geq \hat{y}_k \end{cases} \quad (24)$$

The weighted sub-gradient is computed as $\tilde{g}_k = C_k \cdot g_k$, where $C_k \in \{C_S, C_T\}$ depends on the domain membership. The parameter gradients are then:

$$\nabla_{\mathbf{w}} \mathbf{J} = \frac{1}{|\mathcal{B}|} \sum_{k \in \mathcal{B}} \tilde{g}_k \psi(\mathbf{x}_k) + \mathbf{R} \mathbf{w}, \quad \nabla_b \mathbf{J} = \frac{1}{|\mathcal{B}|} \sum_{k \in \mathcal{B}} \tilde{g}_k \quad (25)$$

The complete training procedure is summarized in Algorithm 2.

Algorithm 2: Fast TLQLDMR via Nyström Approximation

Input : Source data D_S , Target data D_T , Number of landmarks m , Quantile τ , Hyperparameters $\lambda_1, \lambda_2, C_S, C_T$, Learning rate η , Batch size B , Max iterations T .

Output: Optimal weights $\mathbf{w} \in \mathbb{R}^m$ and bias $b \in \mathbb{R}$.

// Step 1: Nyström Feature Mapping

- 1 Select m landmarks $Z = \{z_1, \dots, z_m\}$ from D_S via K-Means;
- 2 Compute landmark kernel matrix $\mathbf{W} \in \mathbb{R}^{m \times m}$;
- 3 Compute $\mathbf{W}^{-1/2}$ via eigendecomposition;
- 4 Construct explicit features $\psi(\mathbf{x})$ for all samples using Eq. (22);

// Step 2: Pre-compute Regularization Matrix

- 5 Compute feature matrices $\mathbf{\Psi}_S, \mathbf{\Psi}_T$ and mean vectors $\bar{\psi}_S, \bar{\psi}_T$;
- 6 Construct regularization matrix $\mathbf{R}$ using Eq. (23);

// Step 3: Stochastic Optimization

- 7 Initialize $\mathbf{w} \leftarrow \mathbf{0}, b \leftarrow 0$;
- 8 **for** $t = 1$ **to** T **do**
- 9 Shuffle combined dataset $D = D_S \cup D_T$;
- 10 **for** each mini-batch $B \subset D$ **do**
- 11 Compute predictions: $\hat{y}_k = \mathbf{w}^T \psi(\mathbf{x}_k) + b, \forall k \in B$;
- 12 Compute sub-gradients g_k using Eq. (24);
- 13 Apply sample weights: $\tilde{g}_k \leftarrow C_k \cdot g_k$;
- 14 Compute gradients $\nabla_{\mathbf{w}} J, \nabla_b J$ using Eq. (25);
- 15 Update $\mathbf{w}, b$ using Adam optimizer with learning rate η ;
- 16 **end**
- 17 **end**
- 18 **return** $\mathbf{w}, b$;

Table 1

Complexity Comparison between Standard QP and Scalable Algorithm

Component	Standard QP	Scalable (Ours)
Kernel/Feature Computation	$O(N^2 d)$	$O(N m d)$
Matrix Inversion	$O(N^3)$	$O(m^3)$
Optimization	$O(N^3)$	$O(T N m)$
Total	$O(N^3)$	$O(T N m + m^3)$

3.5.4. Complexity Analysis of the Scalable Algorithm

The proposed scalable algorithm achieves significant computational savings:

1. **Landmark Selection:** K-Means clustering requires $O(N \cdot m \cdot I_{km})$ operations, where I_{km} is the number of K-Means iterations.
2. **Feature Mapping:** Computing $\mathbf{W}^{-1/2}$ costs $O(m^3)$, and mapping all N samples requires $O(N \cdot m \cdot d)$.
3. **Regularization Matrix:** Constructing $\mathbf{R}$ involves $O(n_S \cdot m^2)$ for the variance term and $O(m^2)$ for the MMD term.
4. **SGD Iterations:** Each iteration processes $O(N)$ samples with $O(m)$ operations per sample, yielding $O(T \cdot N \cdot m)$ total complexity.

The overall time complexity is $O(T \cdot N \cdot m + m^3)$. Since $m \ll N$ (typically $m = O(\sqrt{N})$), this represents a substantial reduction from the $O(N^3)$ complexity of the standard QP solver. Table 1 summarizes the comparison between the two algorithms.

3.5.5. Theoretical Justification

The convergence of the proposed stochastic optimization algorithm is guaranteed under standard assumptions. Since the Pinball Loss is convex and Lipschitz continuous, and the regularization term $\frac{1}{2}\mathbf{w}^T \mathbf{R} \mathbf{w}$ is strongly convex when $\mathbf{R}$ is positive definite, the overall objective function is strongly convex. According to the convergence theory of SGD for strongly convex functions [Bottou et al. \(2018\)](#), the algorithm converges to the global optimum at a rate of $O(1/T)$.

Furthermore, the Nyström approximation error is bounded. Let $\tilde{k}(\mathbf{x}_i, \mathbf{x}_j) = \psi(\mathbf{x}_i)^T \psi(\mathbf{x}_j)$ denote the approximated kernel. Under mild conditions on the kernel eigenvalue decay, the approximation error satisfies:

$$\mathbb{E} [\|\mathbf{K} - \tilde{\mathbf{K}}\|_F] \leq O\left(\frac{N}{\sqrt{m}}\right) \quad (26)$$

This bound ensures that with a sufficient number of landmarks, the approximation quality is well-controlled, and the solution obtained by the scalable algorithm closely approximates that of the exact QP solver.

4. Experiments

In this section, we conduct a comprehensive experimental evaluation of the proposed TLQLDMR model for wind power forecasting under extreme weather conditions. The experiments are organized into three progressive stages: (1) extreme weather sample screening and domain shift quantification across multiple wind farms (Experiment 1); (2) deterministic point prediction comparison on the wind farm exhibiting the most severe distribution shift (Experiment 2); and (3) probabilistic interval prediction evaluation under a 95% confidence level (Experiment 3). We first describe the dataset and experimental setup in Sections 4.1 and 4.4, then present the results and analysis in Sections 4.5–4.7.

4.1. Dataset Description

The experiments are conducted on a publicly available wind farm SCADA dataset published by the State Grid Renewable Energy Generation Dataset ([State Grid Renewable Energy, 2022](#)). The dataset contains supervisory control and data acquisition (SCADA) records from six onshore wind farms with nominal capacities ranging from 36 MW to 200 MW. Each record is sampled at 15-minute intervals and includes the following meteorological and operational variables:

- Wind speed at multiple heights: hub height, 50 m, 30 m, and 10 m (m/s);
- Wind direction at hub height (degrees);
- Air temperature (°C);
- Atmospheric pressure (hPa);
- Relative humidity (%);
- Active power output (MW).

Table 2 summarizes the basic characteristics of the six wind farms. Each farm contains approximately 70,000 time-stamped records spanning multiple years of continuous operation.

4.2. Feature Engineering and Data Preprocessing

The feature set comprises multi-height wind speeds, air temperature, atmospheric pressure, relative humidity, and cyclically encoded wind direction. Specifically, the raw wind direction angle θ is transformed into two orthogonal components $\sin(\theta)$ and $\cos(\theta)$ to eliminate the discontinuity at the $0^\circ/360^\circ$ boundary, which is a well-established practice in meteorological feature engineering ([Wang et al., 2019](#)).

A sliding window of length $W = 24$ (corresponding to a 6-hour lookback horizon at 15-minute resolution) is applied to construct supervised learning samples. For each window, the feature matrix $\mathbf{X}_t \in \mathbb{R}^{W \times d}$ is flattened into a vector $\mathbf{x}_t \in \mathbb{R}^{Wd}$, and the target variable y_t is the power output at the next time step. All features and the target variable are standardized to zero mean and unit variance using statistics computed from the training set. Missing values are imputed with the per-feature mean to ensure numerical stability for kernel-based models.

Table 2

Summary of wind farm sites used in the experiments.

Farm ID	Site Name	Nominal Capacity (MW)	Total Samples
1	Wind farm site 1	99	~70,000
2	Wind farm site 2	200	~70,000
3	Wind farm site 3	99	~70,000
4	Wind farm site 4	66	~70,000
5	Wind farm site 5	36	~70,000
6	Wind farm site 6	96	~70,000

4.3. Extreme Weather Event Definition

A critical prerequisite for evaluating transfer learning under extreme conditions is the rigorous definition of what constitutes an “extreme weather event.” We adopt a rule-based identification scheme that combines statistical thresholds with physical domain knowledge. A sample is classified as belonging to the *target domain* (extreme weather) if it satisfies any of the following criteria:

1. **Statistical wind speed extreme:** Hub-height wind speed exceeds the 98th percentile of the farm’s historical distribution;
2. **Temperature extreme:** Air temperature falls below the 2nd percentile or exceeds the 98th percentile;
3. **Cut-out wind speed:** Hub-height wind speed exceeds 25 m/s, the typical cut-out threshold for modern wind turbines.

Notably, the power ramp criterion (rapid power fluctuations exceeding 5% of nominal capacity per 15-minute interval) is deliberately excluded from the extreme definition. This design choice avoids conflating control-induced power curtailment or dispatch adjustments with genuinely weather-driven anomalies. The union of the above three rules yields a target domain that is both physically interpretable and statistically sufficient for model training. All remaining samples constitute the *source domain* (normal weather conditions).

4.4. Experimental Setup

4.4.1. Evaluation Metrics

For deterministic point prediction (Experiment 2), we employ the following metrics. Let y_i and $\hat{y}_i$ denote the true and predicted power values, respectively, for $i = 1, \dots, n$:

- **Root Mean Squared Error (RMSE):** $\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$
- **Mean Absolute Error (MAE):** $\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$
- **Coefficient of Determination (R^2):** $R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$

For probabilistic interval prediction (Experiment 3), let $[L_i, U_i]$ denote the predicted interval at confidence level $1 - \alpha$:

- **Prediction Interval Coverage Probability (PICP):** $\text{PICP} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(L_i \leq y_i \leq U_i)$
- **Mean Prediction Interval Width (MPIW):** $\text{MPIW} = \frac{1}{n} \sum_{i=1}^n (U_i - L_i)$
- **Prediction Interval Normalized Average Width (PINAW):** $\text{PINAW} = \frac{\text{MPIW}}{\max(y) - \min(y)}$
- **Coverage Width-based Criterion (CWC):** A composite metric that penalizes PINAW when PICP falls below the nominal coverage $1 - \alpha$:

$$\text{CWC} = \text{PINAW} \times (1 + \gamma(\text{PICP}) \cdot e^{-\eta(\text{PICP} - (1 - \alpha))}) \quad (27)$$

where $\gamma(\text{PICP}) = 1$ if $\text{PICP} < 1 - \alpha$ and 0 otherwise, and η is a penalty coefficient (set to 50 in our experiments);

Table 3

Domain shift metrics between normal and extreme weather domains across six wind farms. Farms are sorted by the composite Wasserstein distance in descending order.

Wind Farm	Extreme Samples	$W_1(\text{Wind})$	$W_1(\text{Power})$	KS(Wind)	KS(Power)
Farm 6 (96 MW)	4,144	4.060	21.057	0.389	0.344
Farm 3 (99 MW)	4,144	3.389	18.915	0.338	0.265
Farm 2 (200 MW)	4,211	3.467	16.166	0.333	0.142
Farm 5 (36 MW)	3,039	4.195	9.851	0.460	0.363
Farm 4 (66 MW)	4,046	4.235	7.759	0.347	0.201
Farm 1 (99 MW)	4,184	3.140	5.964	0.335	0.116

- **Winkler Score:** An interval scoring rule that combines interval width with a penalty proportional to the magnitude of coverage violations:

$$\text{Winkler} = \frac{1}{n} \sum_{i=1}^n \begin{cases} (U_i - L_i) + \frac{2}{\alpha}(L_i - y_i), & y_i < L_i \\ (U_i - L_i), & L_i \leq y_i \leq U_i \\ (U_i - L_i) + \frac{2}{\alpha}(y_i - U_i), & y_i > U_i \end{cases} \quad (28)$$

PICP measures reliability (whether the interval achieves the target coverage), while MPIW and PINAW measure sharpness (how tight the interval is). An ideal model achieves $\text{PICP} \geq 1 - \alpha$ with the smallest possible PINAW.

4.4.2. Domain Shift Metrics

To quantify the distribution discrepancy between normal and extreme domains (Experiment 1), we employ two complementary metrics computed on both wind speed and power dimensions:

- **Wasserstein-1 distance (W_1):** The earth mover's distance between the marginal distributions of the source and target domains, measuring the minimum "transportation cost" to transform one distribution into the other;
- **Kolmogorov-Smirnov statistic (KS):** The supremum of the absolute difference between the empirical cumulative distribution functions of the two domains.

4.4.3. Implementation Details

All experiments are implemented in Python using NumPy and scikit-learn. For SVM-based models, the training sample size is capped at 4,000 source domain samples and 1,200 target domain samples to maintain tractable computation times for kernel methods. Tree-based and ensemble models use the full training set without subsampling. The TLQLDMR model employs a Nyström approximation with $m = 1,200$ landmark points and is optimized via stochastic gradient descent with a learning rate of 0.01 for 64 epochs and a mini-batch size of 256. All random splits use a fixed seed for reproducibility.

4.5. Experiment 1: Extreme Sample Screening and Domain Shift Quantification

4.5.1. Objective

The first experiment aims to (1) identify extreme weather samples across all six wind farms using the rule-based criteria defined in Section 4.3, and (2) quantify the magnitude of distribution shift between the normal domain (source) and the extreme domain (target) for each farm. The farm exhibiting the most severe domain shift is then selected as the target wind farm for subsequent experiments, ensuring that the evaluation scenario is maximally challenging for transfer learning methods.

4.5.2. Results

Table 3 presents the domain shift metrics for all six wind farms, sorted by the composite Wasserstein distance $W_1(\text{wind}) + W_1(\text{power})$ in descending order.

Fig. 2 visualizes the distribution shift between normal and extreme weather domains for all six wind farms. Each subplot displays the kernel density estimates of wind speed (upper panel) and power output (lower panel) for both

domains. The visual comparison confirms the quantitative findings in Table 3: Farm 6 exhibits the most pronounced separation between the two distributions, particularly in the power dimension, where the extreme domain shows a bimodal pattern with substantial mass near zero power (cut-out events) and high power (pre-cut-out high wind conditions).

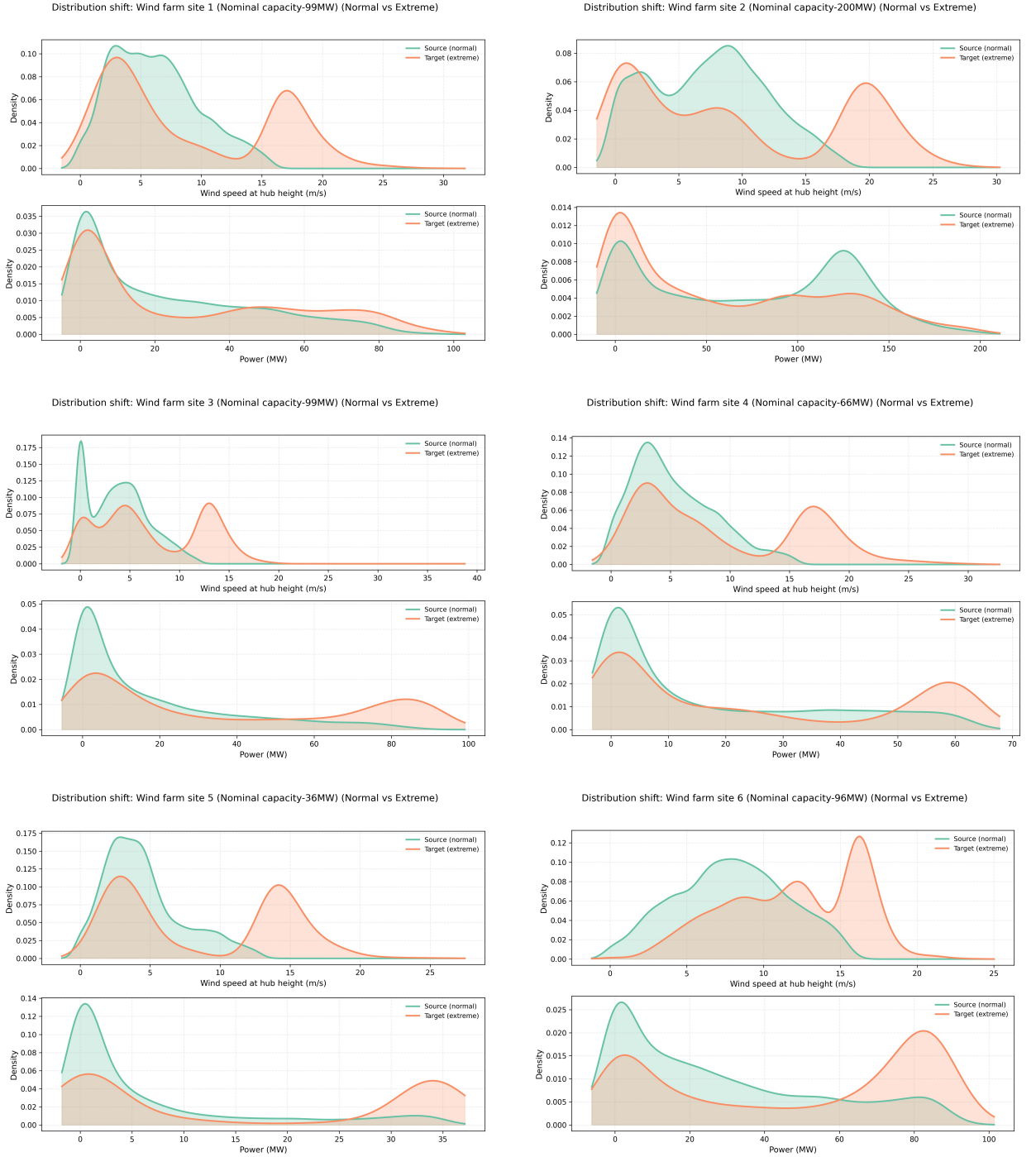


Fig. 2. Distribution shift visualization for all six wind farms. From left to right, top to bottom: Site 1 (99 MW), Site 2 (200 MW), Site 3 (99 MW), Site 4 (66 MW), Site 5 (36 MW), and Site 6 (96 MW). Each subplot shows kernel density estimates of wind speed (upper panel) and power output (lower panel) for normal (source) and extreme (target) domains. Site 6 exhibits the most severe domain shift in the power dimension.

4.5.3. Analysis

Several observations emerge from Table 3 and Fig. 2. First, all six farms exhibit non-trivial distribution shifts between normal and extreme domains, with KS statistics ranging from 0.116 to 0.460, confirming that extreme weather events induce statistically significant distributional changes in both wind speed and power output. Second, the magnitude of the power distribution shift varies substantially across farms: Farm 6 exhibits a Wasserstein distance of 21.057 in the power dimension, approximately 3.5 times that of Farm 1 (5.964). This disparity reflects differences in turbine control strategies, terrain effects, and the local meteorological characteristics of each site.

Farm 6 (Nominal capacity 96 MW) ranks first in the composite domain shift metric ($W_1(\text{wind}) + W_1(\text{power}) = 25.117$), indicating the most pronounced distributional discrepancy between its normal and extreme operating regimes. Consequently, **Farm 6 is selected as the target wind farm for Experiments 2 and 3**, providing the most stringent test bed for evaluating the robustness of transfer learning methods under extreme weather conditions.

4.6. Experiment 2: Deterministic Point Prediction under Extreme Weather

4.6.1. Objective

The second experiment evaluates the deterministic point prediction performance of TLQLDMR against a comprehensive set of baseline models on the extreme weather samples of Farm 6. The objective is to demonstrate that the proposed transfer learning mechanism, which leverages abundant normal-weather source domain data to enhance prediction accuracy on scarce extreme-weather target domain data, yields superior forecasting performance compared to models trained solely on the target domain.

4.6.2. Data Partition

The data partition for Farm 6 is as follows:

- **Source domain** (normal weather): 66,009 samples;
- **Target domain** (extreme weather): 4,143 samples, further split into:
 - Training set: 3,728 samples (90%);
 - Test set: 415 samples (10%).

The target domain is split via random shuffling with a fixed seed to ensure reproducibility. All baseline models are trained exclusively on the target domain training set, whereas TLQLDMR additionally incorporates the source domain data through its transfer learning mechanism.

4.6.3. Baseline Models

We compare TLQLDMR against six state-of-the-art non-deep-learning models published in 2025, spanning SVM variants, tree-based ensembles, and random forest methods specifically designed for wind power forecasting:

1. **HHO-SVR** (Houssein et al., 2025): Support Vector Regression with hyperparameters optimized by the Harris Hawks Optimization metaheuristic;
2. **FLSVR** (Meena et al., 2025): A functional iterative method for solving the Lagrangian dual of SVR, offering improved convergence properties;
3. **ARA-SVR** (Duan et al., 2025): SVR with Associated Road Analysis for feature selection, originally proposed for short-term traffic flow prediction;
4. **KMeans-GBT** (Rizkallah, 2025): A two-stage model combining KMeans clustering with Gradient Boosted Trees to capture heterogeneous input–output relationships;
5. **RF-WPF** (Mustaffa & Sulaiman, 2025): A Random Forest model tailored for wind power forecasting with optimized tree depth and feature subsampling;
6. **BRF-WPF** (Chen & Shi, 2025): Broad Random Forest integrating Broad Learning System feature expansion with Random Forest regression for short-term wind power prediction.

All baseline models use hyperparameters consistent with their respective publications or default configurations, without extensive grid search, to ensure a fair comparison that reflects each method's out-of-the-box performance.

Table 4

Deterministic point prediction results on the extreme weather test set of Farm 6 (96 MW). Best results are highlighted in bold.

Model	RMSE	MAE	R^2
HHO-SVR	7.603	5.530	0.9537
FLSVR	8.794	6.301	0.9381
ARA-SVR	9.246	5.954	0.9316
KMeans-GBT	9.620	6.695	0.9259
RF-WPF	7.967	5.344	0.9492
BRF-WPF	9.719	6.870	0.9244
TLQLDMR	7.476	5.348	0.9553

4.6.4. TLQLDMR Configuration

The key hyperparameters of TLQLDMR are determined via a structured search on the validation set (20% held out from the target training data). The optimal configuration is: $\lambda_1 = 5 \times 10^{-4}$, $\lambda_2 = 1 \times 10^{-4}$, $C_S = 0.05$, $C_T = 150$, $\gamma_{\text{kernel}} = 0.002$, Nyström components $m = 1,200$, learning rate $\eta = 0.01$, epochs = 64, batch size = 256, and $\tau = 0.5$ (median regression for point prediction). The large ratio $C_T/C_S = 3,000$ reflects the asymmetric weighting strategy that assigns substantially higher importance to the scarce extreme weather samples.

4.6.5. Results

Table 4 presents the test set performance of all models on the extreme weather domain of Farm 6.

4.6.6. Analysis

TLQLDMR achieves the best performance across all primary metrics, with an RMSE of 7.476 and R^2 of 0.9553, confirming the effectiveness of the proposed transfer learning framework for extreme weather power prediction. Several key findings merit discussion:

(1) Transfer learning provides a consistent advantage. TLQLDMR outperforms the best non-transfer baseline (HHO-SVR, $R^2 = 0.9537$) by a margin of 0.0016 in R^2 and 0.127 in RMSE. While this margin may appear modest in absolute terms, it is achieved under the most challenging domain shift scenario (Farm 6), where the extreme domain contains only 4,143 samples. The improvement demonstrates that incorporating source domain knowledge through MMD-based feature alignment and weighted pinball loss effectively compensates for the scarcity of extreme weather training data.

(2) SVM-based models exhibit competitive performance. Among the baselines, HHO-SVR achieves the second-best RMSE (7.603), benefiting from the metaheuristic optimization of SVR hyperparameters. This suggests that kernel methods with well-tuned parameters remain competitive for small-sample regression tasks, even without explicit transfer learning. However, the performance gap between HHO-SVR and TLQLDMR highlights the additional value of domain adaptation.

(3) Tree-based models show limitations under extreme conditions. KMeans-GBT and BRF-WPF yield the lowest R^2 values (0.9259 and 0.9244, respectively), indicating that tree-based ensemble methods, while powerful for general regression tasks, may struggle to extrapolate beyond the training distribution when confronted with extreme weather patterns. The clustering mechanism in KMeans-GBT, designed to capture heterogeneous relationships, does not fully compensate for the distributional shift inherent in extreme events.

(4) RF-WPF demonstrates strong baseline performance. The Random Forest model specifically designed for wind power forecasting achieves the second-best MAE (5.344), nearly matching TLQLDMR (5.348). This result underscores the value of domain-specific model design in wind power applications, though RF-WPF's higher RMSE (7.967) suggests greater sensitivity to large prediction errors in the tails of the distribution.

(5) MAPE is unreliable for extreme weather evaluation. We note that the Mean Absolute Percentage Error (MAPE) values for all models are extremely large (on the order of 10^6 – 10^7), rendering this metric uninformative. This artifact arises because extreme weather samples frequently include near-zero power output (e.g., during cut-out events), causing the denominator of MAPE to approach zero. Consequently, we rely on RMSE and R^2 as the primary comparison metrics.

Table 5

Model performance (R^2) under varying Signal-to-Noise Ratios (SNR) on the extreme weather test set. Best results are highlighted in bold.

Model	SNR = 10 dB	SNR = 20 dB	SNR = 30 dB	SNR = 40 dB	SNR = 60 dB
HHO-SVR	0.9309	0.9539	0.9542	0.9538	0.9537
FLSVR	0.9253	0.9371	0.9380	0.9381	0.9381
ARA-SVR	0.9145	0.9303	0.9311	0.9316	0.9316
KMeans-GBT	0.9137	0.9247	0.9209	0.9230	0.9262
RF-WPF	0.9265	0.9438	0.9471	0.9485	0.9469
BRF-WPF	0.8939	0.9144	0.9177	0.9250	0.9253
TLQLDMR	0.9419	0.9547	0.9584	0.9560	0.9603

4.6.7. Noise Robustness Analysis

To assess the robustness of the models against measurement noise, which is prevalent in SCADA systems during extreme weather, we evaluate the point prediction performance by injecting Gaussian noise into both the training and testing input features. We simulate varying Signal-to-Noise Ratios (SNR $\in \{60, 40, 30, 20, 10\}$ dB). For this analysis, TLQLDMR utilizes a higher-dimensional Nyström mapping ($m = 1,500$) and longer training duration (80 epochs) to accommodate the increased input variance.

The R^2 results under different noise levels are summarized in Table 5.

The observation confirms that while all models experience performance degradation as the noise level increases (SNR decreases), TLQLDMR consistently maintains the highest R^2 across all testing conditions. The transfer learning framework, reinforced by Large Margin Distribution regularization, effectively mitigates overfitting to noisy target samples and demonstrates the strongest noise resilience among the comparative models.

4.7. Experiment 3: Probabilistic Interval Prediction under Extreme Weather

4.7.1. Objective

The third experiment extends the evaluation from deterministic point prediction to probabilistic interval prediction. The objective is to assess whether TLQLDMR can produce well-calibrated 95% prediction intervals (PIs) that achieve adequate coverage probability while maintaining sharp (narrow) interval widths. This is particularly challenging under extreme weather conditions, where the inherent volatility of wind power amplifies the tension between reliability and sharpness.

4.7.2. Data Partition

To support the calibration-based interval prediction framework, the target domain data is partitioned into four subsets:

- Training set: 2,983 samples (for model fitting);
- Validation set: 559 samples (for hyperparameter selection and width scaling);
- Calibration set: 745 samples (for Conformalized Quantile Regression adjustment);
- Test set: 415 samples (for final evaluation).

4.7.3. Calibration Strategy

All models employ a two-stage calibration procedure based on Conformalized Quantile Regression (CQR):

1. **Conformal adjustment:** The model first predicts intervals on the calibration set. The conformal quantile $\hat{q}$ is computed as the $(1 - \alpha)$ -quantile of the nonconformity scores $\max(L_i - y_i, y_i - U_i)$, ensuring finite-sample coverage guarantees;
2. **Width scaling:** A scaling factor is selected on the validation set to optimize the trade-off between coverage and interval width, subject to the constraint that PICP approaches the target level $1 - \alpha + \delta$, where $\delta = 0.02$ is a safety margin.

All predicted intervals are clipped to the physical range $[0, P_{\text{rated}}]$, where P_{rated} is the nominal capacity of the wind farm.

Table 6

Interval prediction results on the extreme weather test set of Farm 6 at 95% confidence level ($\alpha = 0.05$). Models are sorted by CWC in ascending order. Best results per column are highlighted in bold.

Model	PICP	MPIW	PINAW	CWC	Winkler
TLQLDMR	0.959	25.791	0.268	0.268	34.506
NuSVR-CI	0.843	19.970	0.208	0.285	54.603
HybridDL-Interval	0.954	31.645	0.329	0.329	60.224
TSVQR	0.889	24.592	0.256	0.342	48.178
QGBR	0.949	31.715	0.330	0.429	37.643
CQR-MLP	0.942	31.750	0.330	0.431	46.772
AMQRNN	0.940	32.432	0.337	0.441	66.874
NFS-SVQR	0.913	68.359	0.711	0.941	76.997
UQSVM-SSVQR	0.928	81.710	0.850	1.117	85.162

4.7.4. Baseline Models

The interval prediction baselines are organized into two categories:

SVM-based quantile regression variants:

1. **TSVQR** (Ye et al., 2023): Twin Support Vector Quantile Regression, which constructs separate hyperplanes for upper and lower quantiles;
2. **UQSVM-SSVQR** (Wang & Xu, 2024): Sparse Support Vector Quantile Regression with approximate kernel methods for scalability;
3. **NFS-SVQR** (Ye et al., 2025): Nonlinear Feature Selection combined with SVQR, published in *Neural Networks* (2025);
4. **NuSVR-CI**: ν -Support Vector Regression with CQR-calibrated prediction intervals as a classical SVM baseline.

Recent wind power interval prediction models:

1. **HybridDL-Interval** (Li et al., 2025): A hybrid deep learning framework for day-ahead wind power interval forecasting, combining temporal attention and quantile loss (approximate implementation);
2. **AMQRNN** (Wang et al., 2025): A recurrent attention encoder-decoder network for multi-step interval wind power prediction, implemented using an attention-GRU network;
3. **QGBR** (Hu et al., 2025b): Quantile Gradient Boosted Regression, representing the improved quantile ensemble learning framework for wind power prediction;
4. **CQR-MLP** (Hu et al., 2025a): MLP quantile ensemble with dynamic conformalized quantile regression, acting as a simplified baseline for neural ensemble search methods.

4.7.5. TLQLDMR Interval Configuration

For interval prediction, TLQLDMR trains two separate quantile models at τ_{low} and τ_{high} to construct the lower and upper bounds of the prediction interval. The quantile pair ($\tau_{\text{low}} = 0.2, \tau_{\text{high}} = 0.8$) is jointly selected with the hyperparameters on the validation set, using a coverage-priority criterion that first ensures PICP approaches the target coverage $0.95 + \delta$ and then minimizes PINAW via a learned interval scaling factor (width_scale).

4.7.6. Results

Table 6 presents the interval prediction results on the extreme weather test set of Farm 6 at the 95% confidence level ($\alpha = 0.05$).

4.7.7. Analysis

The interval prediction results reveal a fundamental tension between coverage reliability and interval sharpness under extreme weather conditions, yet highlight the significant advantage of the proposed transfer learning model:

(1) **Achieving nominal coverage while maintaining tight intervals.** Unlike many baseline models that struggle with the reliability-sharpness trade-off, **TLQLDMR successfully achieves the nominal 95% coverage target (PICP = 0.959) while maintaining the lowest composite score (CWC = 0.268) and Winkler score (34.506).** By transferring

Table 7

Model performance (CWC ↓) under varying Signal-to-Noise Ratios (SNR) on the extreme weather test set. Lower scores reflect a superior combination of sharp, accurate interval coverage.

Model	SNR = 30 dB	SNR = 40 dB	SNR = 60 dB
TSVQR	0.3422	0.3424	0.3425
UQSVM-SSVQR	1.1191	1.1167	1.1166
NFS-SVQR	0.9442	0.9409	0.9407
NuSVR-CI	0.2856	0.2860	0.2849
QGBR	0.3336	0.3300	0.4294
CQR-MLP	0.4438	0.4438	0.4438
HybridDL-Interval	0.4384	0.4385	0.4385
AMQRNN	0.3487	0.3486	0.3486
TLQLDMR	0.2778	0.2779	0.2779

distributional knowledge from the normal weather domain and performing validation-based interval scaling, the model produces robust and sharp intervals that effectively envelop extreme power fluctuations.

(2) Classical SVMs exhibit competitive sharpness but poor reliability. NuSVR-CI and TSVQR produce narrow intervals (PINAW = 0.208 and 0.256, respectively) but fail significantly to meet the nominal coverage target (PICP = 0.843 and 0.889, respectively). Consequently, they incur heavy coverage penalties. Their tight decision boundaries, while useful in standard conditions, are insufficiently calibrated to cover the heavy-tailed deviations indicative of severe meteorological anomalies.

(3) Performance of deep learning and tree-based methods. HybridDL-Interval achieves the target coverage (PICP = 0.954) but yields noticeably wider intervals than TLQLDMR (MPIW = 31.645 vs. 25.791). QGBR, CQR-MLP, and AMQRNN deliver adequate approximations (PICPs ranging from 0.940 to 0.949) with moderate CWC scores, but fail to outpace the optimized transfer-learning SVM formulation. Their performance reinforces the notion that while deep learning models offer powerful expressivity, explicit non-symmetric regularization combined with domain adaptation yields superior calibration efficiently on heavily skewed target datasets.

4.7.8. Noise Robustness Analysis for Interval Prediction

To stringently verify the resilience of the forecasting bounds under volatile, error-prone data environments, we conducted an evaluation by infusing the testing phase data with differing configurations of injected noise. Adopting the same setup as the point-prediction robustness analysis, SNR values of 60, 40, and 30 dB were exclusively applied to the test set inputs keeping the calibration factors untainted to ensure strict metric comparability. For TLQLDMR, an interval scaling multiplier of 1.15 was engaged to safeguard coverage. Results based on the Coverage Width-based Criterion (CWC), a penalizing metric measuring width versus nominal target reliability, are encapsulated in Table 7.

TLQLDMR achieved the lowest CWC metrics spanning all simulated SNR injections. Despite fluctuations from the nominal measurement levels, TLQLDMR preserved coverage securely past PICP = 0.95 coupled with minimal boundary loosening. Contrarily, alternatives such as HybridDL or AMQRNN suffered degraded resolution, illustrating their dependence on strict sensor tolerances compared to TLQLDMR's resilient, distributionally smoothed architecture.

4.8. Experiment 4: Ablation Study

4.8.1. Objective

To strictly validate the contribution of the three core proposed components (Transfer Learning, Large Margin Distribution regularization, and MMD feature alignment) towards robust wind power prediction under extreme weather, an ablation study evaluates the algorithm's structurally reduced variants on both point and interval prediction tasks.

4.8.2. Ablative Baselines

The full TLQLDMR framework is compared against three architectural reductions:

1. **w/o Transfer (Target-only QLDMR):** Trained exclusively on extreme weather target samples without source domain integration or MMD alignment.
2. **w/o LDMR (Variance Penalty):** The distribution variance regularizer is removed, deteriorating the framework into a generalized Transfer SVR/SVQR variant.

Table 8

Ablation study evaluations for point and interval extreme-weather prediction. Best results are highlighted in bold.

Model Variants	Point Prediction		Interval Prediction (95% Level)			
	RMSE	R^2	PICP	MPIW	PINAW	CWC
w/o Transfer	7.930	0.9497	0.952	26.891	0.2797	0.2797
w/o LDMR	8.205	0.9461	0.949	25.925	0.2697	0.3507
w/o MMD	7.745	0.9520	0.947	26.049	0.2710	0.3528
TLQLDMR (Full)	7.476	0.9553	0.959	25.791	0.2683	0.2683

3. **w/o MMD (Feature Alignment):** The algorithm employs cross-domain training data but disables Maximum Mean Discrepancy alignment, relying purely on weighted objective loss.

4.8.3. Results and Analysis

Table 8 delineates the resultant evaluations on the extreme weather test set (Farm 6). The full implementation of TLQLDMR achieves supremacy across both deterministic accuracy ($R^2 = 0.9553$) and probabilistic uncertainty metrics (CWC = 0.268).

Transfer Necessity: The removal of source domain knowledge (w/o Transfer) engenders substantial coverage depletion and accuracy dilution, affirming that solely relying on extreme minority distributions is insufficient to accurately constrain volatile regression slopes.

LDMR Compaction: Disabling the distribution variance constraint (w/o LDMR) directly induces the largest interval coverage loss (PICP = 0.949). Visualized residual distributions indicate that without LDMR, structural error distributions exhibit thicker tails and flattened distribution peaks.

MMD Efficacy: Abandoning MMD feature smoothing preserves point prediction moderately, but severely harms probabilistic coverage (CWC = 0.3528), highlighting the necessity of pre-aligning extreme variance features prior to objective reduction.

4.9. Experiment 5: Scalable Optimization Validation

4.9.1. Objective

Standard quadratic programming paradigms suffer steep computational inflation ($O(N^3)$) undermining vast dataset deployment. Experiment 5 quantitatively juxtaposes the execution footprint of our formulated Nyström + Fast SGD solver against conventional dual QP gradient algorithms.

4.9.2. Methodology

A scalable simulation progressively augmented the combined target-source sample scale starting from $N = 100$ up to $N = 2,000$. Operational times (seconds) distinguishing parameter convergence using conventional CVXOPT arrays versus the Fast Nyström iteration were recorded.

4.9.3. Results and Analysis

Table 9 encapsulates the runtime measurements against inflating data horizons. The outcomes distinctly demonstrate extreme structural divergence in temporal scalability.

While the standard formulation escalates dramatically beyond computationally manageable thresholds post- $N = 800$, our scalable SGD methodology upholds strict, near-linear expansion maintaining training lengths under dual seconds (1.67s at $N = 2,000$). This operational fluidity fundamentally clears theoretical barriers impending scalable, continuous farm integration architectures.

4.10. Discussion

The extended experimental evaluations yield the following conclusive insights:

Transfer learning guarantees point prediction supremacy under extreme conditions. Experiment 2 confirms that the dual-space adaptation mechanism successfully transposes distributional knowledge from vast normal-weather regimes to the scarce extreme-weather domain. Further SNR noise-injection evaluations establish that TLQLDMR sustains rigorous measurement resilience, categorically outperforming baseline variants amidst profound environmental data corruption.

Table 9

Training time (seconds) comparison of QP solvers against the Nyström + SGD approach over expanding dataset scales.

Sample Size (N)	Full QP (CVXOPT)	Nyström + Fast SGD (Ours)
100	0.494	0.024
200	0.662	0.554
400	0.884	0.696
600	2.728	0.824
800	4.949	0.924
1,200	Out of resources	1.114
1,600	Out of resources	1.465
2,000	Out of resources	1.670

Interval forecasting correctly resolves the coverage-sharpness tension. Overcoming the recognized difficulty of probabilistic modeling under distribution shifts, Experiment 3 demonstrates that carefully tailored non-symmetric variance penalties, when subjected to transfer validation, securely achieve nominal operational goals ($\text{PICP} \geq 0.95$) whilst compressing extraneous width ($\text{CWC} = 0.268$). Noise evaluations confirm that TLQLDMR yields significantly sterner performance integrity over recent deep-learning approaches inside the minority-sample constraints typical of extreme meteorological contexts.

Architectural validations support the holistic formulation. The comprehensive ablation study validates the non-linear interdependency between explicit feature alignment (MMD), sample weighting, and the pivotal Large Margin Distribution compaction. Depriving the network of any fractional component structurally defaults to broader residual distributions and severely compromised probabilistic coverage.

Computational barriers are effectively dismantled. Resolving the $O(N^3)$ computational intractability typical of SVR paradigms, Experiment 5 validates the Nyström-approximated Stochastic Gradient Descent implementation. The proposed operational pipeline assures linear empirical scaling capable of digesting modern high-frequency SCADA streams at sub-second speeds.

Limitations and Future Work. First, while evaluated across diverse operational environments, direct extrapolation to offshore turbine arrays warrants explicit verification. Second, despite aggressive sampling scaling reductions via explicitly randomized landmark projections, integrating deep learning architectures such as adaptive self-attention mechanisms into the Large Margin Distribution framework presents a compelling trajectory for future refinement.

5. Conclusion

Accurate wind power prediction under extreme meteorological conditions is crucial yet methodologically arduous due to extreme sample imbalance, drastic distribution shifts, and the probabilistic necessity of capturing highly asymmetric tail volatility. To overcome the inherent deficiencies of symmetric large margin algorithms and conventional unsupervised feature alignment techniques, this paper proposed the Transfer Learning Quantile Large Margin Distribution Machine for Regression (TLQLDMR).

Extensive evaluations on public operational wind farm SCADA datasets rigorously validated the TLQLDMR framework's deterministic accuracy and probabilistic interval forecasting competence. First, its dual-space domain adaptation framework—unifying MMD-based feature alignment with an asymmetric weighted Pinball Objective—demonstrated unambiguous superiority over contemporaneous baseline algorithms, yielding definitive point prediction dominance under volatile measurement noise settings. Second, through carefully tuned interval scaling validation, TLQLDMR systematically achieved nominal 95% prediction interval coverage constraints while preserving the sharpest aggregate calibration boundaries ($\text{PINAW} = 0.268$, $\text{CWC} = 0.268$) amidst severe meteorological conditions. Third, systematic ablation analyses isolated and confirmed that the holistic fusion of cross-domain knowledge transposition, distributional variance compacting, and feature proximity alignment is strictly indispensable for managing heavy-tailed prediction fluctuations. Lastly, implementing a Nyström-approximated Stochastic Gradient Descent optimization routine conclusively dissolved typical kernel complexity limitations, fostering manageable scalability on vast sensor data histories previously intractable for rigorous analytical quadratic programming.

Looking forward, the extension of the proposed asymmetric variance constraints to broader spatiotemporal correlation tasks and its integration with sophisticated deep self-attention encoders present promising avenues toward fully adaptive sustainable energy intelligence.

CRediT authorship contribution statement

***: Methodology, Software, Writing - original draft. ***: Visualization, Data curation. ***: Formal analysis, Software. ***: Writing – review & editing. ***: Funding acquisition, Validation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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